

# RELATIONAL PROGRAMMING LANGUAGE INPUT FORMS

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**TABLE 1.** Primitive Extensional Operations

| Name                     | Input Form          | Publication Form  |
|--------------------------|---------------------|-------------------|
| selection                | $t \text{ sel } x$  | $t \downarrow x$  |
| relative product         | $t \mid u$          | $t \mid u$        |
| construction             | $t, \text{ bar } u$ | $t \bar{,} u$     |
| pair formation           | $x : y$             | $x : y$           |
| union                    | $x \text{ cup } y$  | $x \cup y$        |
| unit set                 | $\text{un } x$      | $\text{un } x$    |
| currying                 | $\text{cur } x$     | $\text{cur } x$   |
| uncurrying               | $\text{unc } x$     | $\text{unc } x$   |
| unique element selection | $\text{theta } x$   | $\theta x$        |
| cardinality              | $\text{size } x$    | $\text{siz e } x$ |
| structure                | $\text{str } t$     | $\text{str } t$   |
| transitive closure       | $t \text{ sup } +$  | $t^+$             |
| empty set                | empty               | $\emptyset$       |

**TABLE 2.** Nonprimitive Extensional Operations: Group 1

| Name                         | Input Form                         | Publication Form        |
|------------------------------|------------------------------------|-------------------------|
| pair list                    | $(x, y)$                           | $(x, y)$                |
| left pair section            | $(x, )$                            | $(x, )$                 |
| right pair section           | $(, y)$                            | $(, y)$                 |
| duplication                  | $\text{DELTA } x$                  | $\Delta x$              |
| membership                   | $x \text{ member } t$              | $x \in t$               |
| nonmembership                | $x \text{ nomem } t$               | $x \notin t$            |
| improper subset              | $s \text{ !subset } t$             | $s \subseteq t$         |
| proper subset                | $s \text{ subset } t$              | $s \subset t$           |
| equality                     | $s = t$                            | $s = t$                 |
| inverse                      | $\text{inv } t, t \text{ sup } -1$ | $\text{cnv } t, t^{-1}$ |
| domain                       | $\text{dom } t$                    | $\text{dom } t$         |
| range                        | $\text{rng } t$                    | $\text{rng } t$         |
| members                      | $\text{mem } t$                    | $\text{mem } t$         |
| left member                  | $\text{Lm } (x, t)$                | $\text{Lm } (x, t)$     |
| right member                 | $\text{Rm } (x, t)$                | $\text{Rm } (x, t)$     |
| member                       | $\text{Mm } (x, t)$                | $\text{Mm } (x, t)$     |
| right univalent              | $\text{run } t$                    | $\text{run } t$         |
| left univalent               | $\text{lun } t$                    | $\text{lun } t$         |
| bi-univalent                 | $\text{bun } t$                    | $\text{bun } t$         |
| initial members              | $\text{init } t$                   | $\text{init } t$        |
| terminal members             | $\text{term } t$                   | $\text{term } t$        |
| reflexive transitive closure | $t \text{ sup } *$                 | $t^*$                   |
| domain restriction           | $p \rightarrow t$                  | $p \rightarrow t$       |
| range restriction            | $t \leftarrow p$                   | $t \leftarrow p$        |
| restriction                  | $t \text{ restr } p$               | $t \uparrow p$          |

**TABLE 3.** Nonprimitive Extensional Operations: Group 2

| Name                       | Input Form | Publication Form       |
|----------------------------|------------|------------------------|
| first member               | alpha t    | $\alpha t$             |
| last member                | omega t    | $\omega t$             |
| initial sequence           | ALPHA t    | $A t$                  |
| final sequence             | OMEGA t    | $\Omega t$             |
| ordered union              | t ; u      | $t ; u$                |
| cons left                  | x cl t     | $x \text{ cl } t$      |
| cons right                 | t cr x     | $t \text{ cr } x$      |
| minimum                    | min s      | $\min s$               |
| maximum                    | max s      | $\max s$               |
| intersection               | s cap t    | $s \cap t$             |
| set difference             | s \ t      | $s \setminus t$        |
| subrange set               | {m .. n}   | $\{m, \dots, n\}$      |
| subrange sequence          | [m .. n]   | $[m, \dots, n]$        |
| apply functional record    | t @ hat x  | $t \hat{\text{ @ }} x$ |
| apply functional structure | t ! x      | $t ! x$                |
| minimize                   | mu t       | $\mu t$                |
| database index             | index x t  | $\text{index } x t$    |
| database select            | select x   | $\text{select } x$     |
| database join              | join x     | $\text{join } x$       |
| array to sequence          | as t       | $\text{as } t$         |
| sequence to array          | sa t       | $\text{sa } t$         |
| seq. to zero-origin array  | sa0 t      | $\text{sa0 } t$        |
| relative product           | rp f t     | $  f t$                |
| relative product inverse   | rpi f t    | $  f t$                |
| array catenation           | t cat u    | $t \text{ cat } u$     |
| relation sort              | rsort s    | $\text{rsort } s$      |
| sort                       | sort s     | $\text{sort } s$       |
| unit image                 | unimg t x  | $\text{unimg } t x$    |
| all                        | all t      | $\text{all } t$        |
| sequence to matrix         | ssm t      | $\text{ssm } t$        |

**TABLE 4.** Primitive Intensional Operations

| Name                 | Input Form            | Publication Form                          |
|----------------------|-----------------------|-------------------------------------------|
| application          | f @ x                 | $f \text{ @ } x$                          |
| image                | img f s               | $\text{img } f s$                         |
| composition          | f . g                 | $f \cdot g, f \cdot g, f \text{ comp } g$ |
| left section         | (x+), (x-), ...       | $(x+), (x-), \dots$                       |
| right section        | (+y), (-y), ...       | $(+y), (-y), \dots$                       |
| paralleling          | f    g                | $f \parallel g$                           |
| isomorphism          | f \$ t                | $f \$ t$                                  |
| formal application   | f @ bar g             | $f \bar{\text{ @ }} g$                    |
| functional condition | (p -> f; g)           | $(p \rightarrow f; g)$                    |
| curry                | curry f               | $\text{curry } f$                         |
| uncurry              | uncurry f             | $\text{uncurry } f$                       |
| filtering            | PHI p (d, r)          | $\Phi p (d, r)$                           |
| iteration            | iter [p -> f]         | $\text{iter } [p \rightarrow f]$          |
| formalization        | + bar, times bar, ... | $\bar{+}, \bar{\times}$                   |
| identity             | Id                    | $\text{Id}$                               |

**TABLE 5.** Nonprimitive Intensional Operations

| Name                 | Input Form      | Publication Form       |
|----------------------|-----------------|------------------------|
| while loop           | while [p, f]    | <b>while</b> [p, f]    |
| reduction            | f red i         | $f \S i$               |
| repeated composition | f sup n         | $f^n$                  |
| value of node        | upsilon f       | $v f$                  |
| operate on form      | phi f           | $\phi f$               |
| operate on data      | delta f         | $\delta f$             |
| image of structure   | PI f            | $\Pi f$                |
| extension            | extend (t, f)   | <b>e xtend</b> (t, f)  |
| restriction          | restrict (s, f) | <b>restrict</b> (s, f) |
| formal negation      | wig p           | $\sim p$               |

**TABLE 6.** Miscellaneous Operations

| Name              | Input Form  | Publication Form |
|-------------------|-------------|------------------|
| sum               | x + y       | $x + y$          |
| difference        | x - y       | $x - y$          |
| product           | x times y   | $x \times y$     |
| quotient          | x divide y  | $x / y$          |
| inequality        | x != y      | $x \neq y$       |
| less              | x < y       | $x < y$          |
| greater           | x > y       | $x > y$          |
| less or equal     | x <= y      | $x \leq y$       |
| greater or equal  | x >= y      | $x \geq y$       |
| conjunction       | x andsign y | $x \wedge y$     |
| disjunction       | x orsign y  | $x \vee y$       |
| negation          | not x       | $\neg x$         |
| cartesian product | s cart t    | $s \times t$     |

**TABLE 7.** Miscellaneous Syntax

| Name                | Input Form         | Publication Form   |
|---------------------|--------------------|--------------------|
| parentheses         | ()                 | ()                 |
| brackets            | []                 | []                 |
| braces              | { }                | { }                |
| sequences           | <x, y, ... >       | <x, y, ... >       |
| identifiers         | a, b', total, etc. | a, b', total, etc. |
| strings             | "abcd"             | "abcd"             |
| data definition     | x == y             | $x \equiv y$       |
| function definition | f x == y           | $f x \equiv y$     |
| output              | display x          | <b>display</b> x   |

**TABLE 8.** Example Program in Publication Form

|                                        |                                  |
|----------------------------------------|----------------------------------|
|                                        | -- F = old file, U = update file |
| H $\equiv$ "Hours"                     | -- Field name for hours worked   |
| u $\equiv$ [H, ] . (F $\hat{=}$ H + U) | -- Updating function             |
| F' $\equiv$ u ; F                      | -- New file                      |
| <b>display</b> F'                      | -- Display result                |

**TABLE 9.** Example Program in Input Form

|                                  |                                  |
|----------------------------------|----------------------------------|
|                                  | -- F = old file, U = update file |
| H == "Hours"                     | -- Field name for hours worked   |
| u == [H, ] . (F @ hat H + bar U) | -- Updating function             |
| F' == u ; bar F                  | -- New file                      |
| <b>display</b> F'                | -- Display result                |